

## Part I

# Fundamentals

# Chapter 1

# The Physical Nature of Very, Very Small Particles and its Impact on their Behavior

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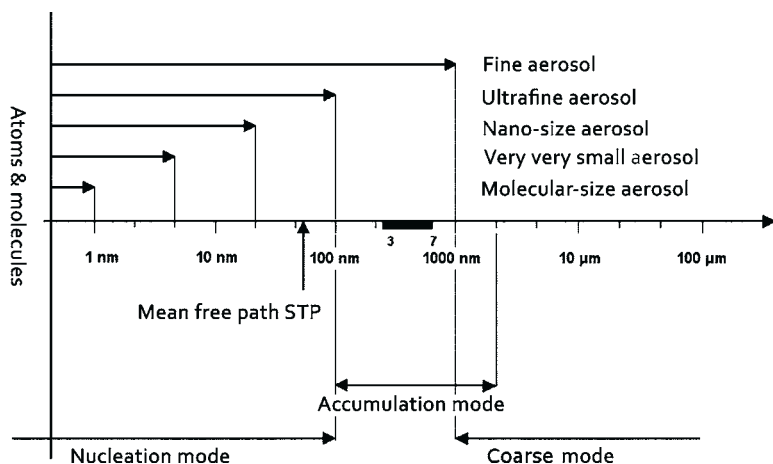
## Chapter Outline

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## 1.1 INTRODUCTION

In aerosol science, particle size is the most important property in governing many aspects of particle behavior. Yet it is a quantity which is not uniquely defined since, depending on the relevant aspect of particle behavior, it may be defined in terms of either its linear geometric dimension or equivalent geometric sizes (i.e. for an equivalent spherical particle) based on the particle volume or surface area, or equivalent sizes for a sphere exhibiting equivalent physical behaviors (e.g. particle aerodynamic diameter and optical diameters). For particles large enough to consider their surfaces or their volumes as continua (larger than about 20 nm), such definitions of particle size are satisfactory. But for much smaller particles, their physical nature—and hence the way in which we think of particle size—needs to be treated differently. In turn, the way in which they interact physically, chemically, and biologically with their surroundings needs to be

\*deceased



**FIGURE 1.1** The aerosol size classes: *coarse mode*, particles larger than about 1  $\mu\text{m}$  mainly produced by diminution processes; *fine aerosol*, particles smaller than about 1  $\mu\text{m}$  mainly built up by nucleation, condensation, and coagulation; *Nucleation mode* and *ultrafine aerosol*, particles smaller than about 100 nm; *Nanosize aerosol*, particles smaller than about 20 nm; *very, very small aerosol*, particles smaller than about 5 nm, particle behavior dominated by surface effects, total number of molecules less than 500; *molecular size aerosol*, particles smaller than about 1 nm, less than 10 molecules in the particle.

reconsidered; molecules from the gas phase statistically associated with the particle may change their physical, chemical, and even their mechanical properties. Figure 1.1 indicates the spectrum of particle sizes ranging from less than 1 nm to 100  $\mu\text{m}$  where the lower end is close to the size of individual molecules and this makes their behavior so distinctly different from that of larger particles.

## 1.2 THE SPECTRUM OF AEROSOL PARTICLE SIZES

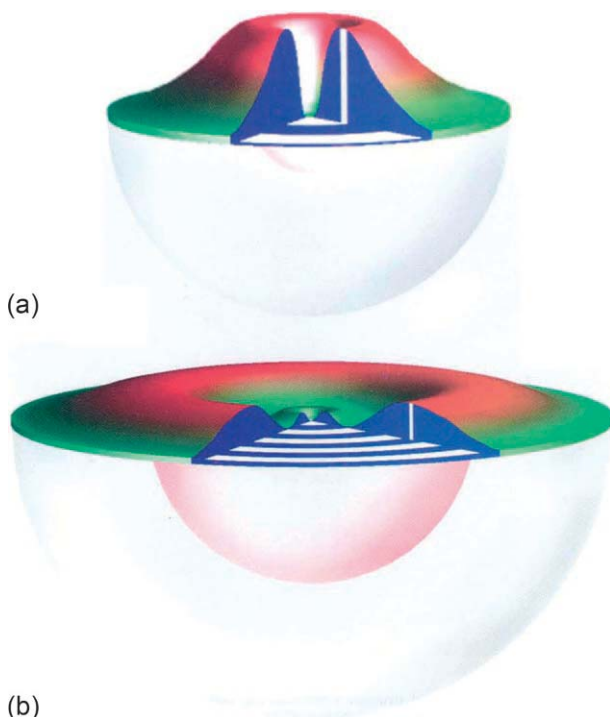
To place into perspective the range of particle sizes pertaining to aerosol science in its widest sense, we first consider how particles have been classified and defined in terms of their size. The most used classification of the particle size spectrum for atmospheric aerosol is the three-mode distribution proposed by Whitby,<sup>24</sup> including the coarse mode, the nucleation mode, and the intermediate accumulation mode (see Figure 1.1). The latter is most prominent in atmospheric aerosols because the mechanisms which remove these particles from the atmosphere are the weakest. On the one hand, for the very fine mode, the lifetime of particles in the atmosphere is limited by the mechanisms of diffusion and coagulation (and finally deposition). On the other hand, for the coarse mode, the lifetime of particles is limited by gravitational sedimentation. The intermediate mode contains particles with sizes ranging from about 100 nm to 1  $\mu\text{m}$ . Of course, the size ranges associated with the three modes cannot be defined exactly. So it is irrelevant whether we define size in terms of particle radii or diameter. For

present purposes, however, we shall use the term diameter since this provides perhaps the most appropriate overall sense of physical dimension.

More generally, beyond discussion of the modes associated with atmospheric aerosol, the size of particles having a diameter less than about 1000 nm may be discussed in terms of how it relates to various physical phenomena. For example, the diameters of particles in the range 0.3–0.7  $\mu\text{m}$  are of the same size as the wavelengths of visible light, and this clearly influences the way in which such particles interact with it. Particles with diameters smaller than about 100 nm are now referred to as “*ultrafine*” (deviating from an earlier definition due to the need to distinguish more subclasses<sup>17</sup>). Further, “*nanometer size*” defines particles with diameters less than about 20 nm; it may be useful to define very, very small particles, smaller than about 5 nm (containing only up to 500 molecules), and finally molecular size for particles with diameters less than 1 nm.

### 1.3 ATOMS AND MOLECULES—CONCEPTS AND DIMENSIONS

At the smaller sizes in this overall spectrum of aerosols, the physics and chemistry—as well as their interactions with biological systems—are of particular interest for those very small particles. Specifically, for particles with dimensions less than 10 nm the question is posed: how can their surfaces be defined? This relates to how we might define the surface of a molecule or an atom. This requires considering the basic physics of simple atoms, e.g. the hydrogen atom. Figure 1.2 illustrates the radial density distribution of the negative charge (electron) outside the atomic nucleus for two states. Typically for the ground state (i.e. the 1 s-state), the radius of the maximum lies approximately at  $r \approx a_0 = 0.053 \text{ nm}$ , with  $a_0$  being the radius of the electron orbit at the lowest quantum state in the Bohr model (in the Bohr model, the electrons circle the molecules held by electrostatic attraction against centrifugal forces). Hence, an effective diameter of the hydrogen atom may be said to be about 0.1 nm. However, as shown in Figure 1.2, the probability distribution is a continuous function (the charge distribution, a probability density can be calculated from the respective solution of the Schrödinger equation, see textbooks of physics, e.g. the Feynman lecture, Vol. III, 1965<sup>7</sup> or Bergmann Schaefer, Vol. 4, 1992<sup>1</sup>) with finite values extending to considerably greater distances. So the diameter of the hydrogen atom cannot be ascribed any exact value valid for all interactions. Although the probability density falls off rapidly for  $r > a_0$ , the atom—in the mathematical sense—has no actual sharply defined outer boundary. Now, if the hydrogen atom is given sufficient energy (e.g. by colliding ions, free electrons, or photons) it may become excited (the electronic charge is elevated to an excitation level, e.g. the 2 s-state). The probability density distribution of the electronic charge is then modified and the maximum of probability density is now at  $r \approx 6a_0 = 0.3 \text{ nm}$  for the 2 s-state. Hence, the



**FIGURE 1.2** Hydrogen atom, s states, spherical symmetry: *Upper-half*, charge distribution of the electron (probability density) over cross-section, vertical axis arbitrary units, radius in units of  $a_0 = 0.053$  nm (first Bohr radius, white bands); *Lower-half*, inner sphere at the maximum of the charge distribution, outer sphere at the density of the outer break off of the charge distribution. (a) The 1 s-state, break off at  $r = 3a_0$  the indicated maximum at  $r = a_0$ ; (b) the 2 s-state, break off at  $r = 10a_0$  the indicated maximum at  $r = 5.5a_0$ .

effective diameter of the atom might now be said to be close to 0.6 nm. Thus, by simply changing the electronic state of the hydrogen atom, its physical size has effectively been increased by a factor of about 6.

In reality, the situation is more complex, however. In general, it is important to underline the point that the definition of atomic or molecular sizes requires applications of quantum mechanics in considering charge distributions for the electronic cloud surrounding the atomic nucleus. Thus, it is the structure of the electron cloud which really defines the atomic size. So, at this level, we cannot speak of atoms or of molecules as solid entities in the way we do of particles in classical aerosol science. Further, the effective atomic diameter is approximately the same for all elements (a few tenths of a nanometer, a range within a factor of three), even, for example, for a heavy element such as uranium whose atom has as many as 92 electrons. This is because the positive charge of the nucleus increases correspondingly, leading to greater attractive forces and

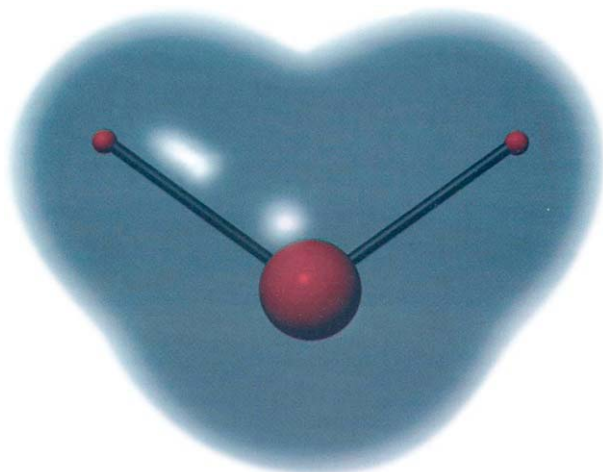
**TABLE 1.1** Selected Bond Lengths in nm

| Compound                      | Bond | Length |
|-------------------------------|------|--------|
| C <sub>2</sub>                | C–C  | 0.1242 |
| CO                            | C–O  | 0.1128 |
| CO <sub>2</sub>               | C–O  | 0.1160 |
| H <sub>2</sub> O              | O–H  | 0.0957 |
| CH <sub>4</sub>               | C–H  | 0.1087 |
| CH <sub>3</sub> ·             | C–H  | 0.108  |
| O <sub>2</sub>                | O–O  | 0.1207 |
| C <sub>6</sub> H <sub>6</sub> | C–C  | 0.1399 |
|                               | C–H  | 0.1101 |

bringing the peak of the respective distribution of the electronic charge closer to the nucleus.

The picture becomes even more complicated when atoms are brought together to build molecules. In order to discuss molecular size it is necessary to consider the distance between the centers of the atoms, the bond lengths, and this in turn depends on how we view the particular structure in which the atoms are brought together. The bond lengths are derived mainly from spectroscopy and are often known accurately (up to four decimal places). [Table 1.1](#) shows the bond lengths for some cases.<sup>4</sup> In a chemical bond, some of the electronic charges are distributed around the (2 or more) positively charged atomic nuclei providing the attractive force. In the water molecule, the single electronic charge of each hydrogen atom is partly associated with the oxygen molecule bringing the total charge of the outer electronic shell of the oxygen close to 10 (the ionic portion of this bond is about one-third).<sup>15</sup>

In addition to these bond lengths, however, it is important to note that the overall size of the entity is also governed by the size of the individual atoms as defined by their electronic charge distributions, and this distribution is complex. Consider, for example, a water molecule made up of two hydrogen atoms and a single oxygen atom. Although the effective size of an isolated single hydrogen atom is of the order of 0.1 nm, its size in the water molecule is considerably smaller. When joined with an oxygen molecule, the respective bond length is about 0.1 nm but the overall size of the water molecule is accordingly larger. The water molecule is intrinsically asymmetric, and the hydrogen atoms are mutually located with respect to the oxygen atom at an angle of 104.45° (instead of 90° due to the p-electron structure) as a consequence of their positive



**FIGURE 1.3**  $\text{H}_2\text{O}$  molecule, the rigid structure symbolizes the well-known distances and angle, the 3D-cloud: the charge distribution of the upper (binding) electrons.

net charge of about  $1/3e$ . The molecule is now an electric dipole. The net result is an asymmetric entity whose size is of the order of 0.2–0.3 nm (see [Figure 1.3](#)).

## 1.4 THE MODEL OF A GAS

The physics of a gas comprising many such molecules is described by the classical kinetic theory. This embodies a somewhat different approach to how molecular sizes are defined, relating to their dimensions (called “*cross-sections*”) which govern how they interact in collisions.

The classical model of a gas involves considering the individual molecules as small, hard spherical particles, in random motion and colliding with one another—and in this process exchanging momentum. This model identifies what is known as an ideal gas. The number of molecules in such a system is vast (there are about  $6 \times 10^{26}$  molecules  $\text{kmol}^{-1}$ , Avogadro’s number) and one mole at standard temperature and pressure (STP) occupies 22.41. The molecular density ( $N$ ) is about  $3 \times 10^{25}$  molecules  $\text{m}^{-3}$ ; from simple geometrical considerations, it may be estimated that the mean distance between centers of molecules is about 3 nm. It is interesting to note that this is only about one order of magnitude greater than the overall geometric dimension of the individual molecules, hence, the density of a gas is about 1/1000 of the density of a liquid or a solid. The thermal speeds of the individual molecules are distributed according to Maxwell–Boltzmann distribution. This means that phenomena arising from the molecular motions depend on parameters of this distribution, e.g. the root mean-square thermal speed  $\left( \sqrt{\langle v^2 \rangle} \right)$ . The value of this velocity

for molecules between collisions (at STP) is high, for example, about  $1700 \text{ ms}^{-1}$  for hydrogen molecules and about  $500 \text{ ms}^{-1}$  for air molecules. It is of interest to consider the distance traveled by a molecule between collisions, the mean free path ( $\lambda$ ). At STP,  $\lambda$  is about 70 nm, which is considerably larger than the average distance between molecules. The concept of  $\lambda$  requires the definition of the collision cross-section denoted by  $\pi \sigma^2$ , with  $\sigma$  the (collision) radius of the molecule, equal to the sum of the radii of the colliding molecules—or if these are of the same kind, simply the diameter. The concept requires the hard sphere model and the Maxwellian velocity distribution. The mean free path multiplied with the collision cross-section  $\lambda \pi \sigma^2$  is simply the volume not containing another molecule on average, the sweep volume. This must be equal to  $1/N$  with  $N$  being the number concentration of molecules. The classical kinetic theory yields for  $\lambda = (\sqrt{2}N\pi\sigma^2)^{-1}$ .

The factor  $1/\sqrt{2}$  comes in by the averaging process and is valid only for an ideal gas (see any textbook on kinetic theory of gases, e.g., Weizel<sup>23</sup>). Since in this expression,  $s$  is the molecular diameter for the equivalent hard sphere in the kinetic theory model and  $\pi\sigma^2$  is the cross-section of the molecule for inter-molecular collisions, the other molecules are considered as points without extension. Note that although  $\sigma/2$  may be closely related to the molecular radius introduced earlier, it is not physically the same since it relates to the behavior of the ideal gas and not specifically to the microscopic properties of each individual molecule. From the kinetic theory of gases, the pressure ( $P$ ) is given by  $P = Nm \langle v^2 \rangle / 3$  in which  $m$  is the molecular mass. (This is equivalent to  $P \propto T$ , with  $T$  the gas temperature.) Since  $P \propto N$  in this equation, the previous equation for  $\lambda$  yields  $P\lambda = \text{constant}$ . Finally, the gas viscosity ( $\eta$ ) is given by  $\eta = 0.217 m (\langle v^2 \rangle)^{1/2} / \pi \sigma^2$ . The collision cross-section as defined by the preceding relations can be evaluated by measuring the macroscopic viscosity. Typical molecular diameters derived in this way are given in Table 1.2.<sup>19</sup>

**TABLE 1.2** Molecular Diameters from Viscosities

| Molecule        | $\sigma$ (nm) | Molecule                        | $\sigma$ (nm) |
|-----------------|---------------|---------------------------------|---------------|
| H               | 0.092         | SO <sub>2</sub>                 | 0.43          |
| O               | 0.13          | CH <sub>4</sub>                 | 0.38          |
| H <sub>2</sub>  | 0.29          | C <sub>2</sub> H <sub>2</sub>   | 0.42          |
| He              | 0.29          | C <sub>6</sub> H <sub>6</sub>   | 0.3           |
| Ne              | 0.28          | Cyclohexane                     | 0.61          |
| Xe              | 0.40          | [NH <sub>4</sub> ] <sup>+</sup> | 0.29          |
| Air             | 0.36          | [OH] <sup>−</sup>               | 0.26          |
| CO <sub>2</sub> | 0.40          |                                 |               |



1.5 PARTICLES AND GAS MOLECULES

The preceding discussion provides the basis for thinking about the physical nature of very, very small aerosol particles. Consider first as a molecule a hypothetical model sphere with diameter  $d=0.5\text{ nm}$  and density  $1000\text{ kg m}^{-3}$ . Its mass is about  $6.5 \times 10^{-26}\text{ kg}$  and it occupies a volume of about  $6.5 \times 10^{-29}\text{ m}^3$ . Its geometrical cross-section is about  $2 \times 10^{-19}\text{ m}^2$ . In terms of atomic mass, the sphere would have a mass of about 40 amu (where  $1\text{ amu} = 1.67 \times 10^{-27}\text{ kg}$ ). So this might be considered as a moderately heavy single molecule. We now consider the broad molecular properties of multi-molecule particles, extending into the nanometer range. Table 1.3 summarizes the cross-section, mass, number of molecules, and the fraction of molecules present at the surface of the particle (where  $d$  is again the particle diameter). The latter is calculated in two ways: (a) using the relatively simple method which can be applied for dense structures and (b) by the method reported by Jimbo.<sup>11</sup> It is seen that they are in reasonable agreement.<sup>17</sup>

In Table 1.3 it is seen that the smaller particles contain only a few molecules practically all at the surface. As particle size increases from 1 to 10 nm, cross-section increases by a factor of 100, and mass and number of molecules by a factor of 1000. Meanwhile, the proportion of molecules at the surface falls from 100% to just 25%. At 20 nm, only about 12% of the molecules are located at the surface of the particle.

Of course, if particles are formed by larger molecules (like large organic molecules), the number of molecules per particle will decrease and their surface fraction will increase. Now consider what happens to the electronic structures of the individual molecules when they come together in these particles. By combining in this way, their electronic structures will mutually interact so that the

| TABLE 1.3 Size Fraction of Molecules at the Surface |                                           |                                  |                        |                                            |     |
|-----------------------------------------------------|-------------------------------------------|----------------------------------|------------------------|--------------------------------------------|-----|
| Size<br>(nm)                                        | Cross-section<br>( $10^{-8}\text{ m}^2$ ) | Mass<br>( $10^{-25}\text{ kg}$ ) | Number of<br>Molecules | Fraction of<br>Molecules<br>at Surface (%) |     |
|                                                     |                                           |                                  |                        | (a)                                        | (b) |
| 0.5                                                 | 0.2                                       | 0.65                             | 1                      | –                                          | –   |
| 1.0                                                 | 0.8                                       | 5.2                              | 8                      | 100                                        | 99  |
| 2.0                                                 | 3.2                                       | 42                               | 64                     | 90                                         | 80  |
| 5.0                                                 | 20                                        | 650                              | 1000                   | 50                                         | 40  |
| 10.0                                                | 80                                        | 5200                             | 8000                   | 25                                         | 20  |
| 20.0                                                | 320                                       | 42,000                           | 64,000                 | 12                                         | –   |

overall chemical behavior of the particles will be different from that of individual molecules—or indeed from free molecules in the gas. They will also be different from the macroscopic bulk state of the substance in question under the same conditions of temperature and pressure. For these very, very small particles the physical situation is therefore unfamiliar.

On the one hand, they are not, and do not behave like, individual molecules; but on the other hand, because they contain so few molecules, they cannot be said to be either liquid or solid. They must therefore be regarded as a state of matter which is distinct in its own right. In this state of matter, in particular, we cannot—unlike for much larger entities—approach particles by thinking of them simply in terms of a solid, well-defined surface. There is no such continuum-based description. Rather, the particle must now be regarded as a complex structure whose properties, and resultant behavior, will depend on the relative positions of the individual molecules in the particle and finally on the structure of the combined electronic charge distribution (probability density).

Consequently, it now becomes apparent that we need to discuss the particle in terms of binding and surface energies of molecules in the particles. Although in the chemistry of materials it is common to express such energy in terms of the units  $\text{kJ mol}^{-1}$ , it is useful here to express it in terms of  $\text{eV molecule}^{-1}$ , where  $1 \text{ kJ mol}^{-1} = 1.036427 \times 10^{-2} \text{ eV molecule}^{-1}$  ( $1 \text{ J} = 6.24151 \times 10^{18} \text{ eV}$ ). Table 1.4 shows the energy associated with some simple inter-molecular chemical bonds.<sup>4</sup> From this it is seen that weak bonds may be characterized by energies in the range of about  $0.1 \text{ eV molecule}^{-1}$  and strong bonds by energies of the order of several  $\text{eV molecule}^{-1}$ . Note here that  $1 \text{ eV}$  corresponds to a

**TABLE 1.4** Bond Strength at 298 K

| Bond                             | $\text{kJ mol}^{-1}$ | $\text{eV molecule}^{-1}$ |
|----------------------------------|----------------------|---------------------------|
| C–C                              | 607                  | 6.29                      |
| Au–Au                            | 225                  | 2.33                      |
| Cu–O                             | 269                  | 2.79                      |
| He–He                            | 4                    | 0.04                      |
| N–O                              | 631                  | 6.53                      |
| H–H                              | 430                  | 4.45                      |
| H–CH <sub>3</sub>                | 439                  | 4.55                      |
| H–OH                             | 498                  | 5.16                      |
| CH <sub>3</sub> –CH <sub>3</sub> | 376                  | 3.90                      |
| O=CO                             | 532                  | 5.51                      |

| TABLE 1.5 Binding and Surface Energies and their Ratios                                                                                   |                      |                       |                      |                       |                                                            |           |                      |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|----------------------|-----------------------|------------------------------------------------------------|-----------|----------------------|
| <i>d</i><br>(nm)                                                                                                                          | <i>E<sub>B</sub></i> |                       | <i>E<sub>S</sub></i> |                       | <i>R</i> = 100 <i>E<sub>S</sub></i> / <i>E<sub>B</sub></i> |           |                      |
|                                                                                                                                           | Au (eV)              | H <sub>2</sub> O (eV) | Au (eV)              | H <sub>2</sub> O (eV) | Au (%)                                                     | Jimbo (%) | H <sub>2</sub> O (%) |
|                                                                                                                                           | 2.3*                 | 0.2*                  | 1*                   | 0.07*                 | –                                                          | –         | –                    |
| 1                                                                                                                                         | 18                   | 1.6                   | 8                    | 0.56                  | 44                                                         | 18.9      | 35                   |
| 2                                                                                                                                         | 147                  | 12.8                  | 56                   | 3.92                  | 38                                                         | 10.0      | 30                   |
| 5                                                                                                                                         | 2300                 | 200                   | 500                  | 35                    | 22                                                         | 4.2       | 17.5                 |
| 10                                                                                                                                        | 18,300               | 1600                  | 2000                 | 140                   | 11                                                         | 2.0       | 8.8                  |
| 20                                                                                                                                        | 147,000              | 12,800                | 7700                 | 540                   | 5                                                          | –         | 4.2                  |
| Note: <i>E<sub>B</sub></i> is the binding energy and <i>E<sub>S</sub></i> is the surface energy.<br>*Hypothetical value for one molecule. |                      |                       |                      |                       |                                                            |           |                      |

temperature of about 11,605 K, so it corresponds to a very large energy compared to the kinetic energy of molecules at STP (at STP  $1/2kT \approx 0.025$  eV). Surface tension for bulk materials may also be expressed in the same way, and it is interesting to note that, for most liquids at STP, it ranges from about 0.01 to 0.07 eV molecule<sup>−1</sup>, assuming a molecular surface density in the bulk medium from about 10<sup>19</sup>–10<sup>20</sup> molecules m<sup>−2</sup>.

In the light of this, we now return to the hypothetical particle introduced earlier and consider gold (Au) and water (H<sub>2</sub>O) particles. Here the binding energy of the particle is estimated by multiplying the single bond energy by the number of molecules; note that this is a crude estimate because the binding energy in a small cluster would be considerably different from that for the bulk material of the same substance. Although the figures shown in Table 1.5 are for the latter, they serve as a useful first approximation for the small cluster more appropriate to very, very small particles.

The surface energies for gold and water are 1000 mN m<sup>−1</sup><sup>3</sup> and 72 mN m<sup>−1</sup><sup>4</sup>. Under the assumption of about 10<sup>19</sup> molecules m<sup>−2</sup>, this figure yields for gold and water about 1 and 0.07 eV molecule<sup>−1</sup>, respectively. The binding energy for the water molecule corresponds roughly to the H-bridge or to a fraction of the evaporation entropy, the binding energy of Au to about 2/3 of the Au–Au bond.

In Table 1.5, the figures in the column marked “Jimbo” are estimates made by Jimbo<sup>11</sup> for metallic particles in general. It is interesting to note that these are markedly different from those calculated for gold.

In this range, the particles may be considered as large molecules—in which case their molecular mean velocities can be estimated by means of kinetic theory. In this way, we find a mean velocity of about 150 m s<sup>−1</sup> for a particle with

**TABLE 1.6** Collisions of Molecules with Particles

| d (nm) | Thermal Velocity (ms <sup>-1</sup> ) | Thermal Sweep Volume (10 <sup>-16</sup> m <sup>3</sup> s <sup>-1</sup> ) | Collisions per ns | Knudsen Number (=2λ/d) |
|--------|--------------------------------------|--------------------------------------------------------------------------|-------------------|------------------------|
| 0.5    | 432                                  | 3.4                                                                      | 9.1               | 280                    |
| 1      | 153                                  | 2.7                                                                      | 7.2               | 140                    |
| 2      | 54                                   | 2.7                                                                      | 7.2               | 70                     |
| 5      | 14                                   | 3.2                                                                      | 8.2               | 28                     |
| 10     | 4.8                                  | 4.2                                                                      | 11.3              | 14                     |
| 20     | 1.7                                  | 5.6                                                                      | 15.1              | 7                      |

$d = 1$  nm, falling to about 5 m/s for  $d = 10$  nm. Associated with such mean random thermal velocity is the equivalent cross-section as the particle sweeps through a given volume per unit time. This thermal sweep volume provides an estimate of the rate of collisions the particle makes with other molecules (at STP, the molecular density is  $2.7 \times 10^{25}$  molecules m<sup>-3</sup>). Table 1.6 summarizes these quantities associated with the motion of the small particle, also including the Knudsen number which is defined as a ratio between the molecular mean free path and the particle radius.

In Table 1.6, the thermal random velocity of the particle falls as the particle gets bigger. On the other hand, the thermal sweep volume is fairly constant. We see that a particle with  $d = 1$  nm has about 7 collisions per ns, rising only to about 10 for  $d = 10$  nm. This is a very high collision frequency, and so this becomes an important consideration when we think, for example, about particles and their reactions with pollutant gas molecules in the atmosphere. Here, therefore, even if the partial pressure of a pollutant gas species is minute, the chance that a very, very small particle will interact with a molecule of that species within 1 s is itself very high. Conversely, the mean free path of the particle between collisions with gas molecules—of any species—is very small; so is the relaxation time of the particles. Thus, from all the preceding calculations, we must think in terms of very large numbers of events happening in very short time intervals.

The particle now considered as a large molecule acquires by collision with gas molecules the same average kinetic energy (equipartition of energy). Since the particle can also rotate it will acquire an average rotational energy accordingly. This rotational energy is stored in an angular velocity of the particle corresponding to its momentum of inertia. Brownian displacement and Brownian rotation play a significant role in the interactions between particles and gas molecules. Table 1.7 summarizes the respective displacements and rotational

| TABLE 1.7 Brownian Motion |                           |                            |                               |                   |                                                 |
|---------------------------|---------------------------|----------------------------|-------------------------------|-------------------|-------------------------------------------------|
|                           | Brownian Displacement     |                            | <i>I</i><br>kg m <sup>2</sup> | Brownian Rotation |                                                 |
| <i>d</i> (nm)             | mm<br>(√1s) <sup>−1</sup> | nm<br>(√1ns) <sup>−1</sup> |                               | Rad in<br>1 ns    | Mean Equatorial<br>Velocity (ms <sup>−1</sup> ) |
| 0.5                       | 5.23                      | 165                        | 1.65 × 10 <sup>−45</sup>      | 1,570             | 393                                             |
| 1                         | 2.64                      | 84                         | 5.24 × 10 <sup>−44</sup>      | 278               | 139                                             |
| 2                         | 1.31                      | 41                         | 1.68 × 10 <sup>−42</sup>      | 49                | 49                                              |
| 5                         | 0.53                      | 17                         | 1.64 × 10 <sup>−40</sup>      | 4.9               | 12                                              |
| 10                        | 0.24                      | 8.3                        | 5.24 × 10 <sup>−39</sup>      | 0.9               | 5                                               |
| 20                        | 0.13                      | 4.1                        | 1.68 × 10 <sup>−37</sup>      | 0.15              | 1.5                                             |

motions for particles of the given sizes assuming they are spheres with uniform density of  $r = 1000 \text{ kg m}^{-3}$ . The momentum of inertia of a sphere is  $I = \pi \rho d^5 / 60$ .

1.6 PARTICLE INTERACTIONS

1.6.1 Coagulation

If 1 g, 1 mg, 1 μg and 1 ng, respectively, of material of density  $1000 \text{ kg m}^{-3}$  is dispersed into  $1 \text{ m}^{-3}$ , then, depending on the size of the individual particles, the number of particles is very large indeed. The half-life of the particles as limited by coagulation can be calculated, given that each collision leads to attachment and coagulation, following the simple equation,  $dn/dt = -Kn^2$ .<sup>10</sup> The results are summarized in Table 1.8.<sup>17</sup>

Here it is seen, for example, that the half-life for  $1 \text{ g m}^{-3}$  of particles with  $d = 1 \text{ nm}$  is of the order of  $1 \text{ μs}$ . So, the process of coagulation for this aerosol is almost instantaneous. Even for  $1 \text{ mg/m}^{-3}$  the half-life is still as short as about 1 ms. At this concentration, the half-life rises to about 1 s when  $d$  is as large as 10 nm. From such considerations, it becomes apparent that coagulation is a very fast process for very, very small particles, so that in practical situations the life-time of such particles is mostly very short.

1.6.2 Homogeneous Nucleation

The conversion of vapor to the particle phase during homogeneous nucleation is very sensitive to the supersaturation ratio ( $S$ ). The particle production rate,  $I$  (in  $\text{m}^{-3} \text{ s}^{-1}$ ), at temperature  $T = 300 \text{ K}$  is given for water by  $I = C_1 \exp [-C_2/T^3(\ln S)^2]$ . The

**TABLE 1.8** Coagulation Constant, Concentration, and Half-life for Very Small Particles

| $d$ (nm) | $K$ ( $10^{-16} \text{ m}^3 \text{ s}^{-1}$ ) | Particle Number for $1 \text{ gm}^{-3}$ | Half-life            |                       |                                 |                       |
|----------|-----------------------------------------------|-----------------------------------------|----------------------|-----------------------|---------------------------------|-----------------------|
|          |                                               |                                         | $1 \text{ gm}^{-3}$  | $1 \text{ mg m}^{-3}$ | $1 \text{ } \mu\text{g m}^{-3}$ | $1 \text{ ng m}^{-3}$ |
| 0.5      | 2.4                                           | $1.5 \times 10^{22}$                    | $3.9 \times 10^{-7}$ | $3.9 \times 10^{-4}$  | $3.9 \times 10^{-1}$            | $3.9 \times 10^2$     |
| 1        | 3.4                                           | $1.9 \times 10^{21}$                    | $2.2 \times 10^{-6}$ | $2.2 \times 10^{-3}$  | $2.2 \times 10^0$               | $2.2 \times 10^3$     |
| 2        | 4.8                                           | $2.4 \times 10^{20}$                    | $1.2 \times 10^{-5}$ | $1.2 \times 10^{-2}$  | $1.2 \times 10^1$               | $1.2 \times 10^4$     |
| 5        | 7.6                                           | $1.5 \times 10^{19}$                    | $1.2 \times 10^{-4}$ | $1.2 \times 10^{-1}$  | $1.2 \times 10^2$               | $1.2 \times 10^5$     |
| 10       | 10.7                                          | $1.9 \times 10^{18}$                    | $7.0 \times 10^{-4}$ | $7.0 \times 10^{-1}$  | $7.0 \times 10^2$               | $7.0 \times 10^5$     |
| 20       | 15.1                                          | $2.4 \times 10^{17}$                    | $3.8 \times 10^{-3}$ | $3.8 \times 10^0$     | $3.8 \times 10^3$               | $3.8 \times 10^6$     |

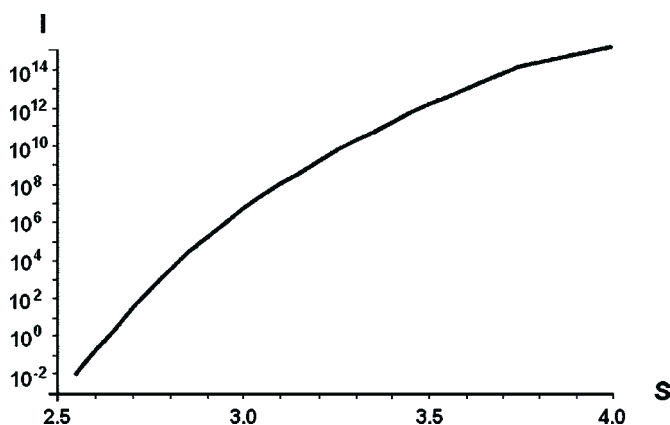


FIGURE 1.4 Nucleation rate ( $I$ ) and supersaturation ratio ( $S$ ) for very small particles.

constants  $C_1$  and  $C_2$  for water are:  $C_1 = 9 \times 10^{32} \text{ s}^{-1} \text{ m}^{-3}$  and  $C_2 = 2.15 \times 10^9 \text{ K}^3$ . Figure 1.4 shows  $I$  as a function of  $S$  for water where the number of particles produced rises by 15 orders of magnitude as  $S$  goes from somewhat less than 3 to about 4. This production rate of primary particles in the nanometer range is so large that, as seen in the preceding discussion, their coagulation must in turn be very fast. In practice, therefore, this means that if there is a process with significant supersaturation (e.g. during expansion), it is not possible to separate the homogeneous nucleation to form nanometer particles from the coagulation of those particles.

### 1.6.3 Adsorption

Now consider the interaction of very small particles with molecules. As already mentioned, molecules of even a minor gaseous constituent have a high probability of interacting with such a particle. Here, the nature of the interaction depends on the adsorption energy of the molecule at the surface. This may be expressed in terms of the residence time of the molecule at the surface.<sup>5</sup> As shown in Table 1.9, the relatively small adsorption energy of  $0.0044 \text{ eV molecule}^{-1}$  (i.e. equivalent to  $100 \text{ cal mol}^{-1}$  in the units usually applied to such situations) corresponds to a residence time of about  $10^{-13} \text{ s}$ . This should be compared with the collision time between molecules which from knowledge of molecular mean random velocities and molecule size may be shown to be of the order of  $10^{-12} \text{ s}$ . However, as the adsorption energy rises to  $0.44 \text{ eV molecule}^{-1}$ , the residence time rises steeply to a value which is about 5–6 orders of magnitude larger than the molecular collision time.<sup>18</sup>

What happens in such molecular collisions with longer residence times? In effect, molecules may be regarded as staying longer in the vicinity of the particle surface. Although this effect has been postulated since the 1930s,<sup>25</sup> only during the 1980s physicochemical techniques have become available to

**TABLE 1.9** Adsorption Energies and Residence Times of Molecules

| Adsorption Energy<br>(kcal mol <sup>-1</sup> ) | Adsorption Energy<br>(eV molecule <sup>-1</sup> ) | Residence Time<br>at the Surface (s) |
|------------------------------------------------|---------------------------------------------------|--------------------------------------|
| 0.1                                            | 0.0044                                            | $1.2 \times 10^{-13}$                |
| 1.5                                            | 0.066                                             | $1.3 \times 10^{-12}$                |
| 3.5                                            | 0.154                                             | $4.0 \times 10^{-11}$                |
| 10                                             | 0.44                                              | $3.2 \times 10^{-6}$                 |
| 20                                             | 0.88                                              | $1.0 \times 10^{-2}$                 |
| 30                                             | 1.32                                              | $4.0 \times 10^0$                    |

measure residence times and reactions as functions of the orientations of molecules. Now, it is known that for asymmetric molecules the reaction between them is strongly dependent on the molecular orientation.<sup>9,14</sup> Importantly, molecules with long residence times near the particle migrate over the surface of the particle and encounter regions of different electronic structure. This means that the probability of finding a configuration favorable for a reaction is increased. This leads to a framework for describing the nature of nanoparticle–molecule interactions.

## 1.7 NANOPARTICLES AS MOLECULAR CLUSTERS

The properties of a molecular cluster may also be thought of in terms of cluster or condensed matter physics. If the entity is a cluster of several molecules, the binding energy will depend on the number of molecules and their orientation.

As mentioned earlier, the system needs to be thought of in terms of its electronic structure. Consider a hypothetical sodium cluster of 20 atoms. Sodium is a metal and each atom contributes one electron to the whole system. So the 20-atom particle has 20 free electrons; however these form a shell-like structure in analogy to the shell structures of atomic nuclei following quantum mechanics. If we look now at the properties of these clusters as a function of the number of sodium atoms, one will find energy extremes for closed shells. Hence, in a random formation process clusters with closed shell structures always even numbered (paired electrons) will be formed more likely than clusters with unpaired electrons. On the other hand, it can be anticipated that clusters with unpaired electrons will be more reactive, hence, their lifetime is reduced and they will be less frequently found. The shell structure has been numerically modeled using a structure of relatively densely packed positively charged nuclei called jellium. The free electrons reside in the potential of this structure. They are now in a field similar to a proton in an atomic nucleus. Hence, a closer analogy exists to the shell model of nuclear physics



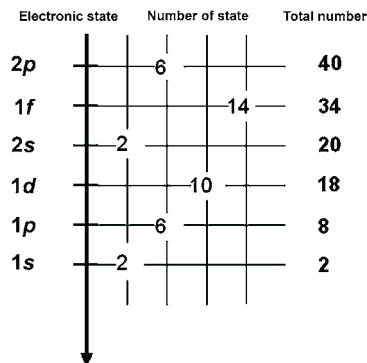


FIGURE 1.5 Energy scheme of electrons in a sodium cluster.

| TABLE 1.10 Binding Energy Per Atom for Na <sup>+</sup> Clusters |               |          |               |
|-----------------------------------------------------------------|---------------|----------|---------------|
| <i>n</i>                                                        | <i>E</i> (eV) | <i>n</i> | <i>E</i> (ev) |
| 2                                                               | 0.49          | 6        | 0.67          |
| 3                                                               | 0.72          | 7        | 0.72          |
| 4                                                               | 0.64          | 8        | 0.70          |
| 5                                                               | 0.71          | 9        | 0.72          |

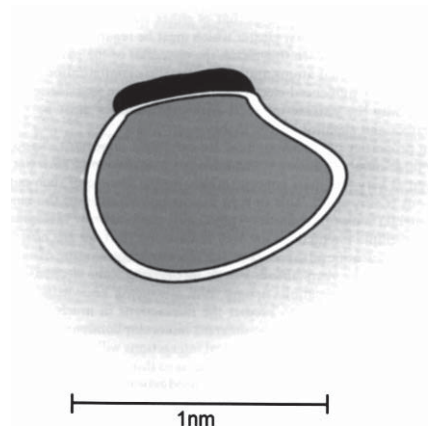
than to that of atoms. Such a model yields the energy levels schematically given in [Figure 1.5](#).<sup>1</sup>

The situation is different for positively charged clusters. Now the binding energy is larger for odd-numbered clusters as recently found, see [Table 1.10](#).<sup>2</sup> The importance of quantum mechanics for describing the (mechanical) properties of nanostructures has been recognized. However, new numerical techniques such as internal coordinate quantum Monte Carlo methods are just yielding first results and progress can be expected in the near future.<sup>13</sup>

Newly formed clusters, particularly cluster aggregates, may have a loose structure containing voids. Such structures can appropriately be described by their fractal dimension. Heat treatment can compact these clusters so they finally get the structure of a solid with a fractal dimension of three.<sup>12,21,22</sup>

### 1.8 AN INTERACTION MODEL FOR NANOMETER-SIZED PARTICLES

Which model is appropriate for describing particle–molecule interactions in the gas phase for an entity with dimensions of a few nanometers or less? We cannot define the surface of the entity as being in any sense a continuum or its volume as being a dense system of atoms or molecules. It is neither liquid nor solid and



**FIGURE 1.6** The very, very small particle. The white band with a width of about 0.05 nm indicates the outer electronic structure of the molecules forming the particle, the cloud around it shows molecules dynamically adsorbed for periods large compared with ideal molecular collision times but small compared to observations (timescale seconds), the black section indicates permanently adsorbed molecules (two-dimensional condensate).

its structure may not be constant with time. The particle itself is very small compared to the mean free path of the gas molecules, and, when the adsorption energy is greater than about  $0.1 \text{ eV molecule}^{-1}$ , the particle will carry adsorbed molecules, which in turn may change the particle properties. The physical picture is portrayed in [Figure 1.6](#), where the so-called nanoparticle phase is surrounded by a gas-phase boundary layer comprising the molecules which have collided with the particle and are migrating over its surface, and where molecular adsorption has taken place at some part of the particle's surface under favorable electronic conditions. The particle is surrounded in its immediate vicinity by a cloud whose molecules are continuously changing within nanoseconds based on the residence time of molecules near the particle. So it can be said that this cloud belongs to the particle.

The particle size has a strong influence on all features of particle–molecule interactions. One example is the Kelvin-equivalent diameter of silver and sodium chloride particles. Such diameters have been measured by Porstendörfer et al.<sup>16</sup> The results are given in [Table 1.11](#). Sodium chloride particles behave in the same way as water droplets of a much larger size and silver particles of a much smaller size. These results show that particle surface and gas reactions are very sensitive to preparation, and small particles behave differently than bulk material of the same substance.

## 1.9 CONCLUDING REMARKS

What may be learned from the basic physics of gases and molecules to identify the main properties of nanometer-sized particles which might have a significant bearing on their behavior, making them distinct from molecules themselves at

| TABLE 1.11 Kelvin-Equivalent Diameters of NaCl and Ag Particles |                     |                                |                     |                                |
|-----------------------------------------------------------------|---------------------|--------------------------------|---------------------|--------------------------------|
| Geometric Diameter                                              |                     | Kelvin-Equivalent Diameter     |                     |                                |
| TEM                                                             | NaCl                |                                | Ag                  |                                |
| d <sub>G</sub> (nm)                                             | d <sub>k</sub> (nm) | d <sub>k</sub> /d <sub>G</sub> | d <sub>k</sub> (nm) | d <sub>k</sub> /d <sub>G</sub> |
| 6                                                               | 14                  | 2.3                            | 3.2                 | 0.53                           |
| 8                                                               | 21                  | 2.6                            | 4.0                 | 0.50                           |
| 12                                                              | 29                  | 2.4                            | 5.0                 | 0.42                           |
| 18                                                              | 40                  | 2.2                            | 7.1                 | 0.39                           |

one end of the size spectrum and from the much larger particles most familiar to aerosol scientists at the other end.

A particle in the size range below about 5 nm comprises a relatively small number of molecules. So it is no longer appropriate to think of the particle as an entity which exhibits the properties of the bulk material from which it is derived. Now we cannot speak of it in terms of a volume or a surface which derives from the assumption that the structure of the particle is a continuum. As has been stated more than once, the nature of the particle, unlike for large particles, cannot be regarded as that of either the liquid or the solid state. The nanoparticle is therefore a molecular cluster which must be regarded as an entirely distinct phase of matter, the nanophase. In this phase, the character of the particle is governed by the properties of the individual atoms and molecules: their configuration, their individual, combined, and mutually influenced electronic states. Therefore, considerations of particle behavior and the way the particle interacts with other entities must be based on both quantum mechanics and classical kinetic theory.

The dynamic properties of the nanophase are extremely important. Nanoparticles are formed in very large numbers during nucleation. But coagulation of such particles occurs very rapidly even at very low mass concentrations. So the lifetime of such particles under normal conditions is very short. This in turn imposes constraints on the kinetics of how such particles interact with other entities—physically, chemically, or biologically.

Currently, only simple interactions are well understood. But we are now beginning to anticipate significant scientific advances based on the very special nature of nanoparticles. It is expected that during the years ahead, kinetic theory, quantum mechanics, and aerosol dynamics, together with applications arising from advancing knowledge of these fundamental disciplines, will enable us to predict the interactions in much more complicated systems. Such basic understanding of the different molecular configurations, their orientation, their Brownian rotation, and their chemical interactions will change our perception

of atmospheric and gas chemistry. This will also force us to think about new approaches in the development of chemical reactors toward the production of new and more advanced materials.

One particularly important aspect of small particles is the way in which they might interact with cells in biological systems. In relation to air pollution, for example, there is increasing concern about the role of small atmospheric particles in the observed increases in disease and mortality in the human populations (e.g. Seaton et al.<sup>20</sup>). In this regard, it seems relevant that Donaldson et al.<sup>6</sup> have demonstrated *in vitro* that some types of very small particles can generate hydroxyl radicals to a far greater extent than larger particles of the same substance. In case such particles are inhaled and deposited in the deep lung, this suggests a mechanism for the generation of harmful oxidants which may be precursors of the types of ill-health observed.

In summary, aerosol scientists and their colleagues in the other disciplines will discover and eventually understand interactions involving nanoparticles which have not yet even been thought about. This will provide great opportunities for advances in aerosol science. As Friedlander<sup>8</sup> remarked at the 1996 Annual Conference of the American Association of Aerosol Research in Orlando, Florida: “. . . *From a scientific point of view it combines the field of aerosol dynamics with aspects of solid state physics and chemistry at the nanometer scale. This will be a very challenging scientific task, both theoretically and experimentally.*”

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